

CHAPTER 5

The d- and f-Block Elements

Transition Elements (d-Block)

1. Zr ($Z = 40$) and Hf ($Z = 72$) have similar atomic and ionic radii because of: (2021)

- Diagonal relationship
- Lanthanoid contraction
- Having similar chemical properties
- Belonging to same group

2. The calculated spin only magnetic moment of Cr^{2+} ion is (2020)

- 4.90 BM
- 5.92 BM
- 2.84 BM
- 3.87 BM

3. Match the following aspects with the respective metal. (2020-Covid)

Aspects		Metal	
A.	The metal which reveals a maximum number of oxidation states	i.	Scandium
B.	The metal although placed in 3d block is considered not as a transition element	ii.	Copper
C.	The metal which does not exhibit variable oxidation states	iii.	Manganese
D.	The metal which in +1 oxidation state in aqueous solution undergoes disproportionation	iv.	Zinc

Select the correct option:

- A-iii B-iv C-i D-ii
- A-iii B-i C-iv D-ii
- A-ii B-iv C-i D-iii
- A-i B-iv C-ii D-iii

4. Match the metal ions given in Column I with the spin magnetic moments of the ions given in Column II and assign the correct code: (2018)

Column-I		Column-II	
A.	Co^{3+}	i.	$\sqrt{8}$ B.M
B.	Cr^{3+}	ii.	$\sqrt{35}$ B.M
C.	Fe^{3+}	iii.	$\sqrt{3}$ B.M
D.	Ni^{2+}	iv.	$\sqrt{24}$ B.M
		v.	$\sqrt{15}$ B.M

- A-iv B-v C-ii D-i
- A-i B-ii C-iii D-iv
- A-iii B-v C-i D-ii
- A-iv B-i C-ii D-iii

5. HgCl_2 and I_2 both when dissolved in water containing I^- ions the pair of species formed is: (2017-Delhi)

- Hg_2I_2 , I^-
- HgI_2 , I_3^-
- HgI_2 , I^-
- HgI_4^{2-} , I_3^-

6. Which of the following processes does not involve oxidation of iron? (2015)

- Decolourisation of blue CuSO_4 solution by iron
- Formation of $\text{Fe}(\text{CO})_5$ from Fe
- Liberation of H_2 from steam by iron at high temperature
- Rusting of iron sheets

7. Magnetic moment 2.84 B.M. is given by: (Atomic numbers, Ni = 28, Ti = 22, Cr = 24, Co = 27) (2015)

- Ti^{3+}
- Cr^{2+}
- Co^{2+}
- Ni^{2+}

8. Because of lanthanoid contraction, which of the following pairs of elements have nearly same atomic radii? (Numbers in the parenthesis are atomic numbers) (2015)

- Zr (40) and Nb (41)
- Zr (40) and Hf (72)
- Zr (40) and Ta (73)
- Ti (22) and Zr (40)

9. Magnetic moment 2.83 BM is given by which of the following ions? (2014)

- (Atomic Number Ti = 22, Cr = 24, Mn = 25, Ni = 28)
- Ni^{2+}
- Cr^{3+}
- Mn^{2+}
- Ti^{3+}



10. Which of the following statements about the interstitial compounds is incorrect? (2013)

a. They retain metallic conductivity
b. They are chemically reactive
c. They are much harder than the pure metal
d. They have higher melting points than the pure metal

Compounds of Transition Elements

11. In the neutral or faintly alkaline medium, KMnO_4 oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is from (2022)

a. +6 to +5 b. +7 to +4
c. +6 to +4 d. +7 to +3

12. Identify the incorrect statement. (2020)

a. The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.
b. Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
c. The oxidation states of chromium in CrO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ are not the same.
d. $\text{Cr}^{2+}(\text{d}^4)$ is a stronger reducing agent than $\text{Fe}^{2+}(\text{d}^6)$ in water.

13. The manganate and permanganate ions are tetrahedral, due to: (2019)

a. The p-bonding involves overlap of p-orbitals of oxygen with d-orbitals of manganese
b. There is no p-bonding
c. The p-bonding involves overlap of p-orbitals of oxygen with p-orbitals of manganese
d. The p-bonding involves overlap of d-orbitals of oxygen with d-orbitals of manganese

14. Which one of the following ions exhibits d-d transition and paramagnetism as well? (2018)

a. CrO_4^{2-} b. $\text{Cr}_2\text{O}_7^{2-}$
c. MnO_4^{2-} d. MnO_4^-

15. Name the gas that can readily decolourise acidified KMnO_4 solution: (2017-Delhi)

a. P_2O_5 b. CO_2
c. SO_2 d. NO_2

16. Which one of the following statements is correct when SO_2 is passed through acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution? (2016-I)

a. SO_2 is reduced b. Green $\text{Cr}_2(\text{SO}_4)_3$ is formed
c. The solution turns blue d. The solution is decolourised

17. Assuming complete ionisation, same moles of which of the following compounds will require the least amount of acidified KMnO_4 for complete oxidation? (2015 Re)

a. $\text{Fe}(\text{NO}_2)_2$ b. FeSO_4
c. FeSO_3 d. FeC_2O_4

Inner Transition Elements (Lanthanoids and Actinoids)

18. Gadolinium has a **low** value of third ionisation enthalpy because of (2022)

a. High basic character b. Small size
c. High exchange enthalpy d. High electronegativity

19. The incorrect statement among the following is: (2021)

a. Most of the trivalent Lanthanoid ions are colorless in the solid state
b. Lanthanoids are good conductors of heat and electricity
c. Actinoids are highly reactive metals, especially when finely divided
d. Actinoid contraction is greater for element to element than Lanthanoid contraction

20. Identify the incorrect statement from the following: (2020-Covid)

a. Lanthanoids reveal only +3 oxidation state.
b. The lanthanoid ions other than the f^0 type and the f^{14} type are all paramagnetic.
c. The overall decreases in atomic and ionic radii from lanthanum to lutetium is called lanthanoid contraction.
d. Zirconium and Hafnium have identical radii of 160 pm and 159 pm, respectively as a consequence of lanthanoid contraction.

21. Match the element in Column-I with that in Column-II. (2020-Covid)

Column-I		Column-II	
A.	Copper	i.	Non-metal
B.	Fluorine	ii.	Transition metal
C.	Silicon	iii.	Lanthanoid
D.	Cerium	iv.	Metalloid

Select the correct option:

a. A-i B-ii C-iii D-iv
b. A-ii B-iv C-i D-iii
c. A-ii B-i C-iv D-iii
d. A-iv B-iii C-i D-ii

22. The reason for greater range of oxidation states in actinoids is attributed to: (2017-Delhi)

a. 4f and 5d levels being close in energies
b. The radioactive nature of actinoids
c. Actinoid contraction
d. 5f, 6d and 7s levels having comparable energies

23. Which one of the following statements related to lanthanons is incorrect? (2016-II)

a. All the lanthanons are much more reactive than aluminium
b. $\text{Ce}(\text{+4})$ solutions are widely used as oxidizing agent in volumetric analysis
c. Europium shows +2 oxidation state.
d. The basicity decreases as the ionic radius decreases from Pr to Lu.

The d- and f-Block Elements

24. The electronic configurations of Eu (Atomic Number 63) Gd (Atomic Number 64) and Tb (Atomic Number 65) are: (2016-I)
- $[\text{Xe}]4f^7 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^9 6s^2$
 - $[\text{Xe}]4f^7 6s^2$, $[\text{Xe}]4f^8 6s^2$ and $[\text{Xe}]4f^8 5d^1 6s^2$
 - $[\text{Xe}]4f^6 5d^1 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^9 5d^1 6s^2$
 - $[\text{Xe}]4f^6 5d^1 6s^2$, $[\text{Xe}]4f^7 5d^1 6s^2$ and $[\text{Xe}]4f^8 5d^1 6s^2$
25. Gadolinium belongs to 4f series. Its atomic number is 64. Which of the following is the correct electronic configuration of gadolinium? (2015 Re)
- $[\text{Xe}]4f^6 5d^2 6s^2$
 - $[\text{Xe}]4f^8 6d^2$
 - $[\text{Xe}]4f^9 5s^1$
 - $[\text{Xe}]4f^7 5d^1 6s^2$
26. Reason of lanthanoid contraction is: (2014)
- Increasing nuclear charge
 - Decreasing nuclear charge
 - Decreasing screening effect
 - Negligible screening effect of 'f' orbitals
27. Which of the following lanthanoid ions is diamagnetic? (Atomic Number Ce = 58, Sm = 62, Eu = 63, Yb = 70) (2013)
- Ce^{2+}
 - Sm^{2+}
 - Eu^{2+}
 - Yb^{2+}

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	a	a	a	d	b	d	b	a	b	b	c	a	c	c	b	b
18	19	20	21	22	23	24	25	26	27							
c	a	a	c	d	a	a	d	d	d							